

CHAPTER FOUR

Project Cost Management





What is Cost and Project Cost Management?

- **Cost** is a resource sacrificed or fore-gone to achieve a specific objective or something given up in exchange
 - Costs are usually measured in monetary units like birr, dollar, etc.
- **Project cost management** includes the processes **required to ensure** that the project is completed within an approved budget
 - Project managers must make sure their **projects are well defined**, have **accurate time** and **cost estimates** and have a **realistic budget** that they were **involved in approving**



The Importance of Project Cost Management

- Software projects have a poor track record for meeting budget goals
- The CHAOS studies found the average **cost overrun** (the additional percentage or dollar amount by which actual costs exceed estimates) ranged from 18% in 1994 to 43% in 2002; other studies found overruns to be 33-34%



Reasons for Cost Overruns

- Not emphasizing the importance of realistic project cost estimates from the outset
 - Many of the original cost estimates for IT projects are low to begin with and based on very **unclear project requirements**
- Many software professionals think preparing cost estimates is a job **for accountants when** in fact it is a very demanding and important skill that project managers need to acquire
- Many software projects involve new technology or business processes which involve untested products and inherent risks.



Project Cost Management Processes

1. **Resource planning:** determining **what** resources and **quantities** of them should be used
2. **Cost estimating:** developing an estimate of the **costs** and **resources** needed to complete a project
3. **Cost budgeting:** allocating the overall cost estimate to **individual work items** to establish a baseline for measuring performance
4. **Cost control:** controlling changes to the project budget



Basic Principles of Cost Management

- ◎ **Life cycle costing** is estimating the cost of a project over its entire life
- ◎ **Cash flow analysis** is determining the estimated annual costs and benefits for a project
- ◎ **Sunk cost** are retrospective (past) costs that have already been incurred and cannot be recovered
 - Sunk costs should not be a criteria in project selection



Basic Principles of Cost Management

Practical Example

- Imagine you spend \$10,000 developing a new app, but before launch, a competitor releases a superior version that makes yours obsolete.

Sunk Cost: The \$10,000 you already spent.

Rational Decision: You should abandon your app. Continuing to spend another \$5,000 on marketing and launch is irrational because you know the app will fail, regardless of the initial \$10,000.

Fallacy: Continuing to spend the \$5,000 because "we've already put in \$10,000, we can't quit now!"



Cost of Software Defects

Phase of Software Development	Relative Cost to Repair Defects
Requirements and Analysis	1×
Coding and Unit Test	5×
Integration and System Test	10×
Beta Test	15×
Post-Product Release	30×
NB: × is a normalized unit of cost and can be expressed in dollars, person-hours, et.	

- It is much more **cost-effective** to spend money on defining **user requirements** and doing **early testing on IT projects** than to wait for problems to appear after implementation
 - If it would cost \$1,000 to repair a software defect in the requirements and analysis phase but it would cost \$30,000 to fix it in the post-product release phase



Resource Planning

- ◎ The nature of the **project and the organization** will affect resource planning
- ◎ Some questions to consider:
 - How **difficult** will it be to do specific tasks on the project?
 - Is there anything **unique in this project's scope** statement that will affect resources?
 - What is the **organization's history** in doing similar tasks?
 - Does the organization have or can it acquire the people, equipment, and materials that are capable and available for performing the work?



Cost Estimating

- An important output of **project cost management** is a cost estimate
- There are several types of cost estimates, and tools and techniques to help create them
- It is also important to develop a cost management plan that describes how cost variances will be managed on the project



Cost Estimation Tools and Techniques

- ◎ 3 basic tools and techniques for cost estimates:
 - **Analogous or top-down:** use the actual **cost of a previous, similar project** as the basis for the new estimate.
 - **Bottom-up:** estimate individual work items and sum them to get a total estimate
 - **Parametric:** use project characteristics in a mathematical model to estimate costs



Typical Problems with Cost Estimates

- ① Developing an estimate for a large software project is a complex task requiring a significant amount of effort. Remember that estimates are done at various stages of the project
- ① Many people doing estimates have little experience doing them. Try to provide training and mentoring
- ① People have a bias toward underestimation. Review estimates and ask important questions to make sure estimates are not biased
- ① Management wants a number for a bid, not a real estimate. Project managers must **negotiate with project sponsors to create realistic cost estimates**



Cost Budgeting

- Cost budget involves allocating the project cost estimate to individual work items and providing a cost baseline
 - For example, in the Business Systems Replacement project, there was a total purchased costs estimate for FY97 of \$600,000 and another \$1.2 million for Information Services and Technology.



Cost Control

- Project cost control includes
 - monitoring cost performance
 - ensuring that only appropriate project changes are included in a revised cost baseline
 - informing project stakeholders of authorized changes to the project that will affect costs
- **Earned value management** is an important tool for cost control



Earned Value Management (EVM)





Earned Value Management (EVM)

- EVM is a **project performance measurement technique** that integrates **scope**, **time**, and **cost** data
- Given a baseline (original plan plus approved changes), you can determine **how well the project is meeting its goals**
- You must enter actual information periodically to use EVM.



Earned Value Management

- Earned Value Analysis is an **industry standard** way to:
 - measure a **project's progress**,
 - **forecast** its completion date and final cost, and
 - provide **schedule** and **budget variances** along the way.
- By integrating three measurements, it provides consistent, numerical indicators with which you can **evaluate** and **compare** projects.



EMV enables



1. Knowing where you are on schedule?
2. Knowing where you are on budget?
3. Knowing where you are on work accomplished?



Earned Value Management Terms

- ⦿ **Planned value (PV)**, formerly called the **budgeted cost of work scheduled (BCWS)**, also called **budget**, is that portion of the approved total cost estimate planned to be spent on an activity during a given period
- ⦿ **Actual cost (AC)**, formerly called **actual cost of work performed (ACWP)**, is the total of direct and indirect costs earned in accomplishing work on an activity during a given period
- ⦿ **Earned value (EV)**, formerly called the **budgeted cost of work performed (BCWP)**, is the percentage of work actually completed multiplied by the planned value



Earned Value Formulas

TERM	FORMULA
Earned Value	$EV = BAC \times \text{Percent Completed}$
Cost Variance	$CV = EV - AC$
Schedule Variance	$SV = EV - PV$
Cost Performance Index	$CPI = EV/AC$
Schedule Performance Index	$SPI = EV/PV$

To estimate what **it will cost to complete a project or how long it will take based on performance to date**, divide the budgeted cost or time by the appropriate index.



Rules of Thumb for EVA Numbers

- ⦿ **Negative numbers** for cost and schedule variance indicate **problems** in those areas.
 - The project is costing more than planned or taking longer than planned
 - Zero variance shows that the project is running according to the plan
- ⦿ CPI and SPI **> 1.0** indicate **exceptional performance**
- ⦿ CPI and SPI **< 1.0** indicate **poor performance**
 - If CPI or SPI = 1, it shows that the project is performing according to its plan



Example

- Suppose you have a software project which is planned to be completed in 9 months with a budget of Birr 900,000. After a month, 10% of the project is completed at a total expense of Birr 100,000, but the planned completion was 15%.
 - Given:
 - Budget At Complete (BAC) = Birr 900,000
 - AC = Birr 100,000



Compute the following:

- a) PV
- b) EV
- c) CV - interpretation
- d) SV – interpretation
- e) CPI - interpretation
- f) SPI - interpretation
- g) Forecast -Budget at complete
- h) Forecast - Time at complete
- i) Overall project's traffic light status



Solution

- a) Planned Value = Planned Completion (%) * BAC
= 15% * Birr 900,000
= **Birr 135,000**
- b) Earned Value = Percent Completed (%) * BAC
= 10% * Birr 900,000
= **Birr 90,000**



...solution

- $CV = EV - AC$
 $= 90,000 - 100,000$
 $= -10,000$

The project is costing more than planned because CV is less than zero.

- $SV = EV - PV$
 $= 90,000 - 135,000$
 $= -45,000$

The project is taking longer than planned because SV is less than zero.



...solution

- $CPI = EV / AC$
 $= 90,000 / 100,000$
 $= 0.90$
- $SPI = EV / PV$
 $= 90,000 / 135,000$
 $= 0.67$

It shows Poor Performance because CPI and SPI are less than one.



Cost Forecasting

- If the project continues at the current performance, what is the true cost of the project?
- Estimate at Complete
 - = Budget at Complete (BAC) / CPI
 - = Birr 900,000 / 0.90 = **Birr 1,000,000**
- At the end of the project, the total project cost will be Birr 1,000,000



Time Forecasting

- If the project continues at the current performance, what is the true time of the project?
- Estimate at Complete
 - = Original Time Estimate / SPI
 - = 9 months / 0.67 = **13.43 months**
- The project will be completed by the end of 13.34 months.



Establish Ranges to Guide Traffic Light Status

- Traffic light status is useful in conveying overall project's status with one color
- Establish objective SPI and CPI ranges to determine the true project color.
- Average of CPI & SPI i.e. $(CPI + SPI) / 2$

Green

[1.0 - 0.95]

Yellow

[0.94 - 0.85]

Red

[0.84 - 0]

Good

Warning

Bad



Overall traffic light status

- Therefore, for the above example, overall project's traffic light status is

$$= (CPI+SPI)/2$$

$$= (0.90+0.67)/2$$

$$= 0.78 \quad \text{Bad}$$

many thanks

